KRISH JOSHI

AI Evaluation Engineer | Agent Trajectory Forensics & Deterministic Test Harnesses

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PROFESSIONAL SUMMARY

Analytical **AI Evaluation Engineer** and 3rd-year CS undergraduate specializing in multi-turn agent trajectory forensics and containerized benchmark design. Promoted to Senior Quality Reviewer (R1) during a completed Handshake AI engagement, auditing frontier model tool-execution traces across simulated personal assistant ecosystems to catch action hallucinations, invalid JSON argument schemas, and grounding failures. Experienced in authoring deterministic Docker evaluation environments (Harbor/Terminal-Bench standards) with two-tier PyTest harnesses and SHA-256 state assertions to prevent agent false passes.

CORE COMPETENCIES & TECHNICAL SKILLS

- **AI Evaluation & Benchmarking:** Forensic Execution Trace Audits, Action Hallucination Detection, Pairwise Reward Modeling (RLHF), Benchmark Task Design (Harbor / Terminal-Bench format), pass@k Calibration
- **Verification & Test Engineering:** Two-Tier PyTest Test Harnesses, Ephemeral Docker Sandboxing, CTRF Schema Reporting, Cryptographic SHA-256 State Assertions, IEEE-754 Precision Bounds
- **Trajectory Auditing & Diagnostics:** Tool-Calling Schema Adherence, Multi-Turn Interaction Transcripts, HTTP Error Recovery (401/403/429 loops), Knowledge Graph Grounding, Disentangling Platform vs. Model Failures
- **Languages, Tooling & Systems:** Python (PyTest, NumPy), Bash / POSIX CLI, Docker Containers, Git/GitHub CI/CD, JSONL Logging, REST APIs, Linux Security Controls

PROFESSIONAL EXPERIENCE

Handshake AI — Frontier Model Alignment & Evaluation
July 2026 – October 2026 (Completed)

AI Evaluation Specialist & Senior Quality Reviewer (R1) — Remote Contractor — *100% Quality Score | SQS 4.5–5.0 Calibration*

- **Forensic Execution Trace Interrogation (Project Octahedron):** Audited multi-turn conversational agent transcripts by inspecting raw JSON execution traces, identifying critical *Action Hallucinations* where models declared task completion without executing corresponding backend API calls (e.g. confirming calendar events or messaging without events.insert or messages.send).
- **Pairwise Reward Modeling (RLHF):** Graded model arms comparatively to train reward models, evaluating trade-offs across instruction adherence, parameter schema validity, linguistic register (multilingual Gujarati & English), and safety.
- **Platform vs. Model Error Diagnostics:** Diagnosed transient sandbox/harness infrastructure drops from genuine model capability flaws (invalid JSON payloads, hallucinated tool schemas), ensuring models were never penalized for platform timeouts.
- **Knowledge Graph Grounding:** Fact-checked model assertions against multi-modal persona databases (email archives, calendar schedules, contact directories) to eliminate invented dates, names, and ungrounded statements.
- **Senior Reviewer Standing:** Promoted to R1 Senior Reviewer with an uninterrupted 100% approval rate and 4.5–5.0/5.0 Standard Quality Score (SQS) across 32 audited tasks (27+ hours logged).

Handshake AI — Containerized Benchmark Engineering (Project Dynamo)
July 2026 – September 2026 (Completed)

Benchmark Task Author & Harness Engineer — Remote Contractor — *Terminal-Bench & Harbor Standards*

- **Containerized Evaluation Environments:** Engineered 10 isolated Docker evaluation tasks in Harbor / Terminal-Bench format to stress-test frontier coding agents on repository refactoring, GPU kernel optimizations, and state reconciliation.
- **Two-Tier Deterministic Verification:** Built two-tier PyTest evaluation suites separating 2-second smoke probes from deep integration assertions, using cryptographic SHA-256 hashes to guarantee 0% false pass rates.

FEATURED ENGINEERING & EVALUATION PROJECTS

AgentTrace — Forensic Agent Observatory & Interceptor
github.com/krish9-gif/ai-systems-eval-portfolio

Python, JSONL, Model Context Protocol (MCP), PyTest, CTRF Schema | Open Source (2026 – Present)

- Engineered an automated trajectory auditor that intercepts *Action Hallucinations* (ghost tool calls, masked 401/403/429 loops) by comparing conversational claims against backend execution receipts; verified across 22 test suites in 0.03s.
- Implemented calibrated SQS (0.0–5.0) grading rubrics, automated CTRF test reporting, and path-traversal sandboxing (assert_safe_path).

agent-mmu & SkillSpectre — Rust Context Memory & Security Scanner
github.com/krish9-gif/agent-mmu

Native Rust, FastMCP, Stdio JSON-RPC, Static Analysis, AST | Open Source (2026 – Present)

- Engineered a zero-dependency Rust memory management unit bounding agent memory in O(1) space (≤8,500 tokens) with >88% KV cache hit rates and SHA-256 inode action anchoring.
- Integrated Layer 6 **SkillSpectre** static security scanner detecting prompt injection, credential exfiltration, destructive execution, and tool poisoning in <15ms across 480+ files (13/13 tests pass).

AgentFixer — Autonomous Agent Runtime Shield & Token Sieve
github.com/krish9-gif/agentfixer

Python, Zero-Dependency Stdlib, AST Analysis, CI/CD Actions | Open Source (2026 – Present)

- Architected a zero-dependency runtime shield for coding agents that compresses terminal test logs by 90%–98% to eliminate context window exhaustion and prompt instruction amnesia.
- Engineered automated fuzzy-patch failure repair and false-pass interceptors to prevent ungrounded agent task approvals.

EDUCATION & PROFESSIONAL CERTIFICATIONS

Bachelor of Technology (B.Tech) in Computer Science & Engineering
July 2024 – June 2028 (Expected)

Institute of Advanced Research (IAR) — Gandhinagar, Gujarat, India — **Current Status: 3rd-Year Undergraduate**

- **Google Cybersecurity Professional Certificate:** Linux security controls, asset defense, vulnerability triage, Python automation.
- **NVIDIA Introduction to Networking:** Modern network architectures, low-latency communication, distributed systems interconnects.